

DNA Replication vs Transcription vs Translation:

Where Errors Change Everything

Medicinal & Scientific
- M.L.S

DNA REPLICATION



Purpose

- Copies entire DNA
- Ensures continuity
- Identical DNA strands

Location

- Nucleus or Cytoplasm
- Origin of replication

Template Used

- Both DNA strands
- Semi-conservative

Main Enzyme

- DNA Polymerase
- Helicase, Ligase

End Product

- Two identical DNAs
- Old & new strands

Timing

- S Phase of cycle
- Once per division

TRANSCRIPTION



Purpose

- DNA to RNA
- Gene expression
- Genetic message

Location

- Nucleus or Cytoplasm
- At specific genes

Template Used

- One DNA strand
- Antisense strand

Main Enzyme

- RNA Polymerase
- No primer needed

End Product

- mRNA molecule
- Processed RNA

Timing

- As needed
- By gene activity

TRANSLATION



Purpose

- mRNA to Protein
- Forms proteins
- Functional output

Location

- Ribosomes
- Cytoplasm, ER

Template Used

- mRNA codons
- Amino acids

Main Components

- Ribosome & tRNA
- Peptide bonds

End Product

- Polypeptide chain
- Functional protein

Timing

- Ongoing process
- Stops at Stop Codon